STARS

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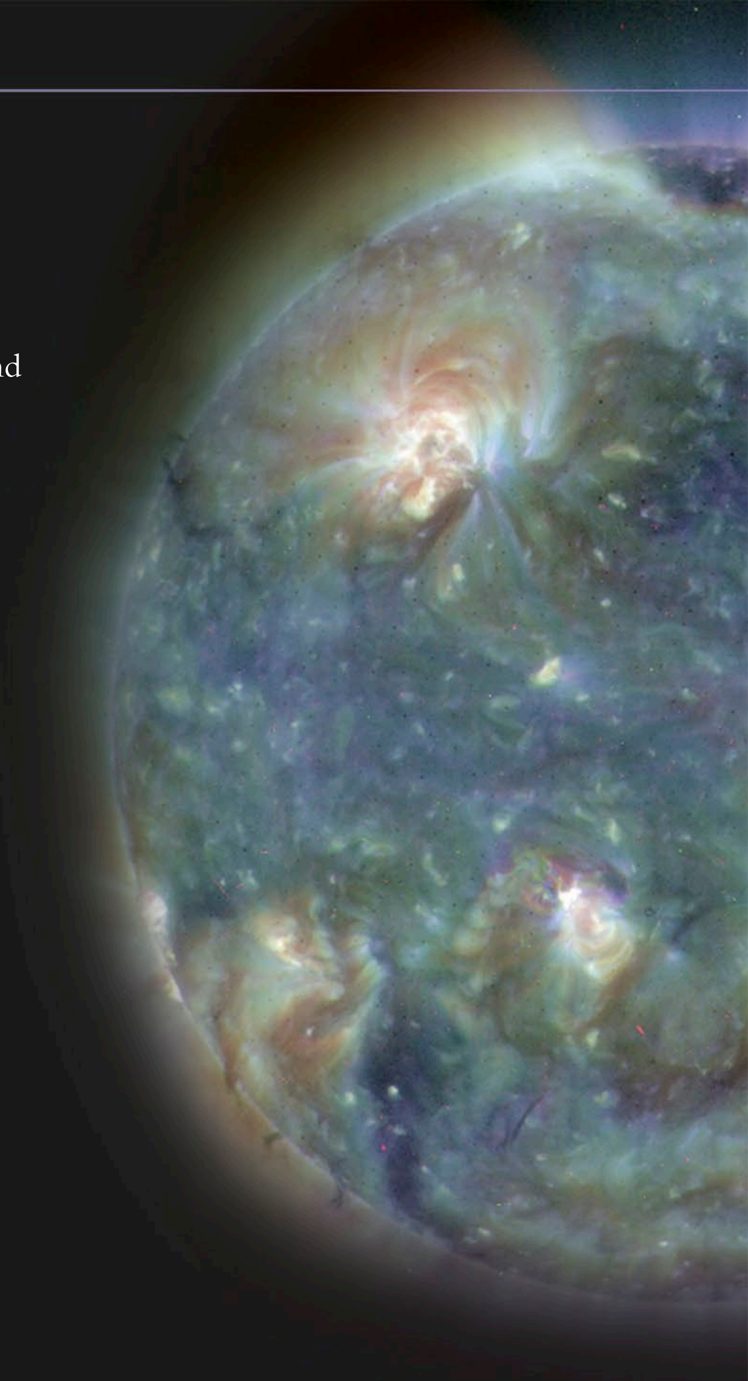
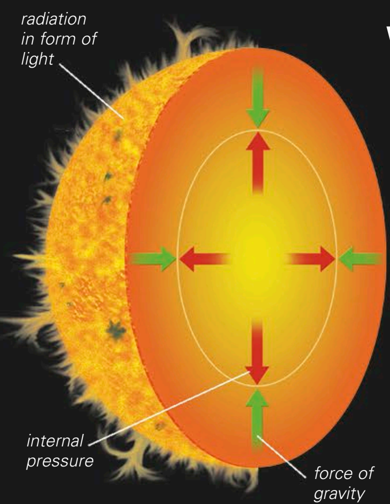


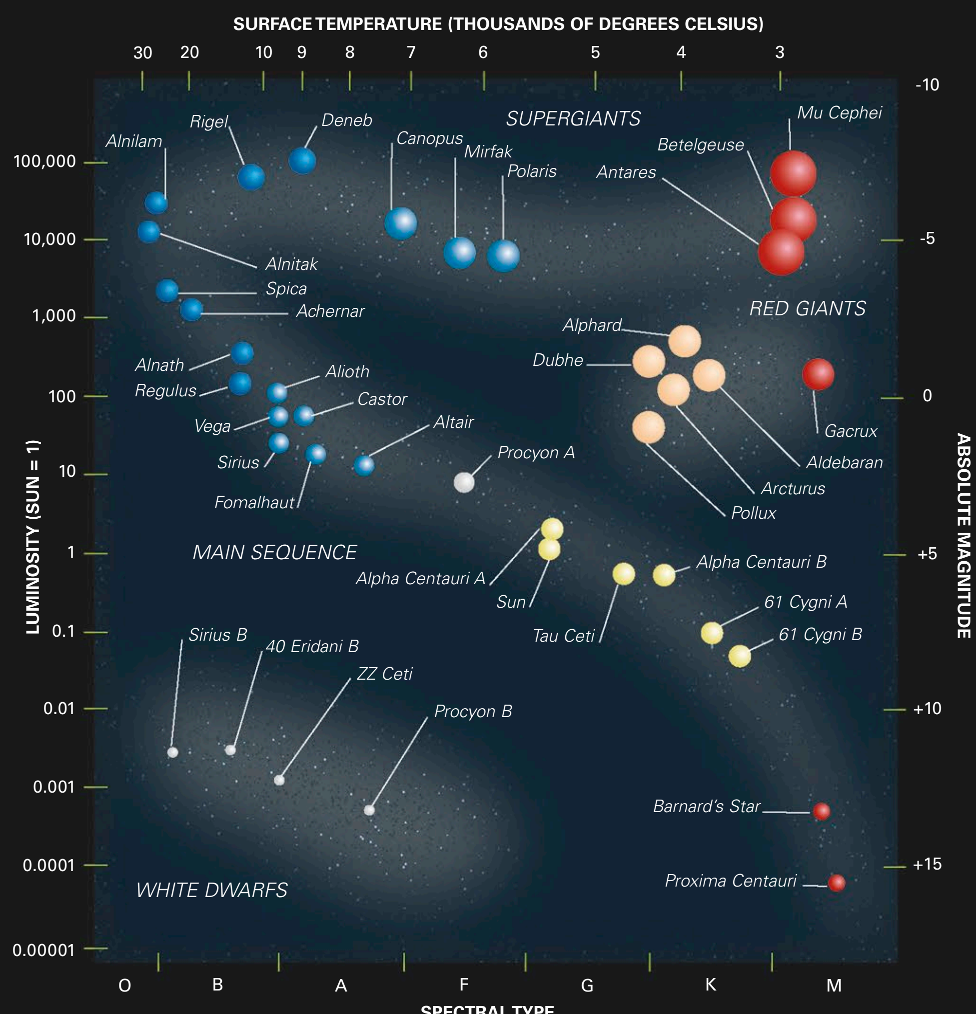
STARS ARE MASSIVE gaseous bodies that generate energy by nuclear reactions and shine because of this energy source. The mass of a star determines its properties—such as luminosity, temperature, and size—and its evolution over time. Throughout its life, a star achieves equilibrium by balancing its internal pressure against gravity.



PRESSURE BALANCE
The state and behavior of any star, at any stage in its evolution, are dictated by the balance between its internal pressure and its gravitational force.

WHAT IS A STAR?

A collapsing cloud of interstellar matter becomes a star when the pressure and temperature at its center become so high that nuclear reactions start (see pp.238–239). A star converts the hydrogen in its core into helium, releasing energy that escapes through the star's body and radiates out into space. The pressure of the escaping energy would blow the star apart if it were not for the force of gravity acting in opposition. When these forces are in equilibrium, the star is stable, but a shift in the balance will change the star's state. Stars fall within a relatively narrow mass range, since nuclear reactions cannot be sustained below about 0.08 solar masses and in excess of about 100 solar masses stars become unstable. A star's life cycle, as well as its potential age, is directly linked to its mass. High-mass stars burn their fuel at higher rates and live much shorter lives than low-mass stars.



THE H–R DIAGRAM

Named after the Danish and American astronomers Ejnar Hertzsprung and Henry Russell, the Hertzsprung–Russell (H–R) diagram graphically illustrates the relationship between the luminosity, surface temperature, and radius of stars. The astronomers' independent studies had revealed that a star's color and spectral type are indications of its temperature. When the temperature of stars was plotted against their luminosity, it was noticed that stars did not fall randomly, but tended to be grouped. Most stars lie on the main sequence, a curved diagonal band stretching across the diagram. Star radius increases diagonally from bottom left to top right. Protostars evolve onto the main sequence as they reduce in radius and increase in temperature. On the main sequence, stars remain at their most stable before evolving into red giants or supergiants, moving to the right of the diagram as their radius increases and their temperature falls. White dwarfs are at the bottom left with small radii and high temperatures.

IMPORTANT DIAGRAM
The H–R diagram is the most important diagram in astronomy. It illustrates the state of a star throughout its life. Distinct groupings represent different stellar stages, and few stars are found outside these groups, since they spend little time migrating.